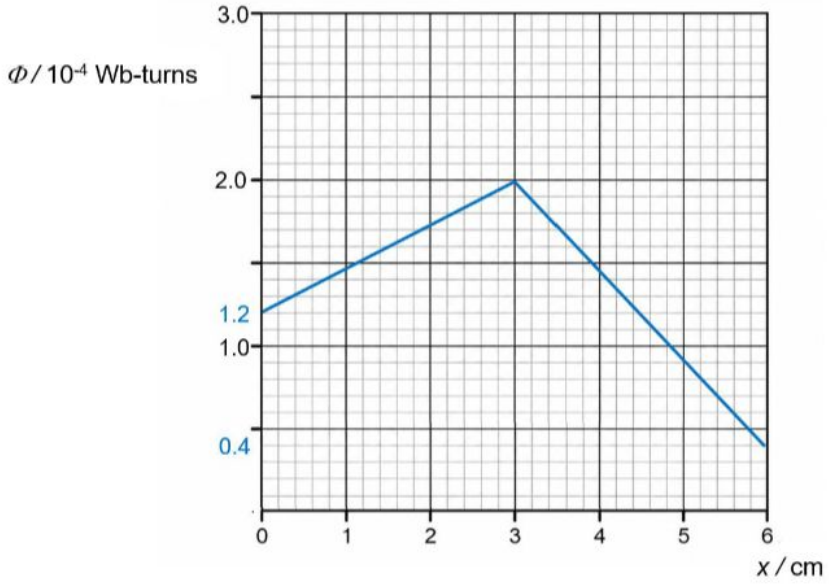
6(a)(i)	From Faraday's equation, Emf = - change in flux linkage / time taken, $\Rightarrow dt$ = - change in flux linkage / emf $= -(+8.0 \times 10^{-5}) / (-5.0 \times 10^{-4})$ $= 0.16$ s Speed = displacement / dt $= 0.03 / 0.16 = 0.1875 = 0.19$ m s^{-1}	[1] [1]
(ii)	Increase in flux linkage = $N \times (5 - 3) \times 10^{-6} = 8.0 \times 10^{-5}$ Wb-turns $\Rightarrow N = 40$ turns	[1]
(iii)	Since the coil moves at constant speed, $\mathcal{E}$ = constant and $\mathcal{E} = -d\Phi / dt$ thus $-d\Phi / dt$ = constant, the Φ vs t graph should have constant gradient. From P to x = 3.0 cm, At P, magnetic flux = 3×10^{-6} Wb (given in part (ii)), thus magnetic flux linkage = N(magnetic flux) = 40 $(3 \times 10^{-6}) = 1.2 \times 10^{-4}$ Wb turns The magnetic flux linkage increases by 8.0×10^{-5} Wb-turns from P to x = 3.0 cm (given in part (i)). Thus final magnetic flux linkage at x = 3.0 cm is $1.2 \times 10^{-4} + 0.8 \times 10^{-4}$ $= 2.0 \times 10^{-4}$ Wb-turns. This is a positive gradient to show an increase in flux linkage. From x = 3.0 cm to x = 6.0 cm, Since $\mathcal{E}$ is doubled and changed in sign from -5×10^{-4} V to $+10 \times 10^{-4}$ V and $\mathcal{E} = -d\Phi / dt$, the gradient in this region should be doubled and negative to show a decrease in flux linkage.  This mark is for those who read ai) and noticed -5×10^{-4} V n $+10 \times 10^{-4}$ V in emf graph • inverted V shape	[1]

	• gradient of 2nd segment = twice of 1st segment [both hv equal x-axis so check the change in Φ is twice (16 small sqs) in segment 2 compared to segment 1 (8 small sqs)] • the change in Φ for 1st segment must be the value in ai) i.e. 8.0 x 10^{-5} Wb-turns This mark is for students who used the values in aii) to calculate the initial Φ • correct initial flux linkage at 1.2 x 10^{-4} Wb-turns	[1]
(b)	By **Faraday's law**, ring **cuts the magnetic flux/field lines** and an **e.m.f. is induced** in the ring. Induced e.m.f. causes **induced/eddy currents to form** in the ring. Either, By **Lenz's law,** the **induced currents flow** in such a direction so as to oppose the change in flux/ flux cutting the ring, So an resistive/opposing force is induced/produced on the ring to oppose its motion. **Or** Currents in the ring **dissipates thermal energy,** And this thermal energy **comes from/is converted from the mechanical energy** of the oscillations.	[1] [1] [1] [1] [1] [1]